

## Systematic Review

# A Review of Subjective Indoor Air Quality Assessment in Non-Residential Buildings: Current Trends and Recommendations

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## Abstract

Survey campaigns in non-residential buildings show that occupants are often dissatisfied with the indoor environmental quality (IEQ), including the indoor air quality (IAQ) conditions. Occupant-centric controls (OCCs) have the potential to improve occupants' satisfaction with IAQ and thermal comfort. Currently, applications of OCC systems with IAQ perceptions are limited due to a lack of a suitable modelling approach to predict occupants' subjective IAQ assessment. In addition, a comprehensive overview of possible confounding variables for subjective IAQ in non-residential buildings is missing. This paper presents a systematic review of 46 papers on subjective IAQ assessments during field investigations in non-residential buildings. The following characteristics of the studies are examined: (1) the study context, (2) study and survey type, (3) dataset and sample size, (4) subjective IAQ assessment scales, (5) analysis and modelling techniques, and (6) associated variables. The review identified 46 different assessment scales and 20 different analysis techniques, respectively, indicating a lack of uniformity across the studies. The vast majority of studies were conducted in classrooms or offices. Other non-residential buildings, such as hospitals and sports halls, were underrepresented. Moreover, most of the studies failed to elaborate on the choice of a statistical technique and to report on the required sample size, compromising the validity of the statistical results. Furthermore, the review highlighted the limited scope of the subjective IAQ assessment analysis, with half of the reviewed studies investigating no more than four different variables. Lastly, only three of the reviewed papers focused on determining an accurate predictive model for subjective IAQ assessment.

**Keywords:** subjective IAQ assessment; IAQ satisfaction; systematic review; non-residential; field studies



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## 1. Introduction

### 1.1. Background and Context

Multiple occupant survey campaigns in non-residential buildings have shown that occupants are often dissatisfied with the indoor environmental quality (IEQ), including the indoor air quality (IAQ) conditions [1–4]. However, good IEQ and IAQ conditions are crucial to optimise employees' and scholars' working [5,6] and learning [7–9] performance [10]. The American Society of Heating, Refrigerating, and Air-Conditioning Engineers (ASHRAE) defines an acceptable IAQ as “air in which there are no known

contaminants at harmful concentrations as determined by cognisant authorities and with which a substantial majority (80 % or more) of the people exposed do not express dissatisfaction” [11]. This definition shows that evaluating indoor air quality involves meeting both objective and subjective requirements. The former involves using measurement devices to determine whether air pollutant concentrations exceed threshold values that could harm occupants’ health. The latter assesses occupants’ satisfaction with the IAQ by gathering feedback directly from occupants on how they perceive the IAQ in a room, for example, through surveys.

Heating, ventilation, and air conditioning (HVAC) control strategies aim to provide IEQ conditions that meet both objective and subjective requirements while using the least amount of energy. However, current control strategies often fail to meet the subjective requirements, as they cannot maximise occupants’ satisfaction with the IEQ due to the use of fixed and conservative setpoints that are not representative of the occupants’ actual IEQ preferences [12,13]. Occupant-centric control (OCC) is a control strategy that is responsive to occupants’ presence, preferences, and behaviour, and it aims at maximising occupants’ satisfaction with IEQ by using Information and Communications Technology (e.g., robust data acquisition systems and wireless communication) [12–15]. Multiple OCC strategies implementing occupants’ thermal preferences have been defined [16–19], which can improve occupants’ thermal comfort by 29.1% and reduce HVAC energy consumption by 22% [14]. However, meeting occupants’ thermal preferences does not guarantee occupant satisfaction with the IAQ. Consequently, complaints about stuffy air and unpleasant odours can still occur. Therefore, a control system that responds to occupants’ subjective IAQ assessments is needed. This system would take into account occupants’ personal perceptions and feelings regarding IAQ, such as satisfaction levels and perception of odours or stuffiness. Currently, IAQ-driven control of ventilation systems is solely based on objective setpoints (e.g., CO<sub>2</sub> concentration, relative humidity) but neglects occupants’ subjective IAQ assessment. Consequently, such control systems can provide indoor environments that meet objective IAQ requirements (e.g., by ensuring that concentrations of IAQ pollutants remain below threshold values), but they cannot guarantee occupant satisfaction with the IAQ. Yang et al. [20] tested the impact of a demand-controlled ventilation (DCV) system with temperature-dependent CO<sub>2</sub> setpoints on perceived air quality, but the controller did not include occupants’ IAQ preferences. Occupant-centric ventilation control systems, integrating occupants’ IAQ preferences, have been investigated only rarely. Carbonare et al. [21] defined and tested an occupant-centric ventilation controller for residential buildings based on humidity, CO<sub>2</sub> concentration and occupant feedback. The self-learning controller responded to the occupant’s feedback (i.e., wanting a higher or lower air exchange) [21].

The lack of OCC systems integrating occupants’ preferred IAQ conditions is caused by the absence of a modelling approach to predict IAQ preferences from occupant feedback in real-life conditions. The most well-known model to predict occupants’ satisfaction with the IAQ was established by Fanger in 1988 [22]. Fanger introduced the concepts of olf and decipol to quantify the emission rate of a pollution source in relation to a reference source (i.e., the emission of one standard person) and the perceived air pollution considering dilution by ventilation or perceived air quality (PAQ), respectively [22]. Based on experiments with 168 human subjects, a relationship was established between the predicted level of dissatisfaction with the IAQ and the decipol or PAQ [22]. These concepts are the basis for standards for determining ventilation flow rates in buildings dominantly polluted by human bioeffluents (e.g., EN 16789-1 [23]). The predicted level of dissatisfaction, from which the PAQ can be inferred, can be calculated from people’s assessment of the acceptability of the air quality scale [24]. Fanger’s olf-decipol model is beneficial to predict the satisfaction with IAQ for a group of people. However, catering the predictions to unique

occupant IAQ preferences is not possible. Moreover, the model was fitted to experimental data and not field measurement data.

Further experimental studies have quantified the impact of environmental variables, such as air enthalpy and air velocity, on humans' subjective IAQ assessment [25–27]. As experimental studies are typically conducted in a climate chamber, they often overlook contextual variables present in real-life (e.g., interior design and proximity to traffic) that could influence occupants' IAQ perception. In field studies, data on the effects of variables on occupants' subjective IAQ assessment are spread out; consequently, a comprehensive overview of the influential variables, which can form the basis for a suitable modelling approach for predicting occupants' IAQ preferences, is missing. The current literature review aims to provide insights into various factors, including those related to IEQ, as well as personal and contextual variables, that influence subjective assessment of IAQ in non-residential buildings.

In the literature, the term PAQ is often used in a general way to refer to humans' assessments of air quality perception, including odour intensity, air freshness, and air dryness. However, in the current review, PAQ will be exclusively used to refer to the value originating from the predicted percentage of dissatisfaction with IAQ, which is determined from the evaluation of the acceptability of the air quality. In this study, we apply the term "subjective IAQ assessment" as a container term relating to evaluations of occupants' perception or satisfaction with the IAQ, including PAQ. In this study, an evaluation is categorised as a "subjective IAQ assessment" if it contains two aspects: (1) it originates directly from occupants' opinion (e.g., through the completion of a survey), and (2) it evaluates the whole or parts of IAQ (e.g., IAQ satisfaction or odour perception). The first aspect highlights that occupants' perception or satisfaction with the IAQ should come directly from occupant feedback instead of assuming satisfaction when measured IEQ variables meet objective requirements (e.g., CO<sub>2</sub> concentrations staying below a threshold value). The second aspect emphasises the broad range of occupants' opinions regarding the IAQ that are considered. They are not limited to the overall IAQ acceptability but may also include evaluations of IAQ satisfaction or specific perceptions, such as odour, freshness, or air humidity.

### *1.2. Novelty of the Current Review*

Existing literature reviews related to or partly tackling occupants' subjective IAQ assessment are summarised in Table 1. Many of these review papers focus narrowly on a limited set of environmental variables (e.g., relative humidity, CO<sub>2</sub> concentration, and temperature) or specific aspects of the ventilation system (e.g., ventilation rate or ventilation filters). However, they often neglect the influence of personal or contextual variables in their review, despite evidence from previous studies showing that occupants' satisfaction with IEQ [28], behaviour [29], and thermal perception [30] are significantly influenced by personal and contextual variables. Therefore, the current review expands upon this by systematically investigating the impact of IEQ-related, personal, and contextual variables on subjective IAQ assessment.

Furthermore, Table 1 highlights that current review papers either concentrate on residential buildings [31], mix residential and non-residential contexts [32–35], or include both field and experimental data [36–39]. Torriani et al. [39] performed a systematic literature review on the assessment methods for subjective IAQ in office buildings. Their review combined both field studies and experimental studies in which an office environment was recreated. Their literature review did not discuss influencing factors for subjective IAQ assessments. Another review by Torriani et al. [38] explored the role of personal and contextual variables in IAQ perception within office environments but excluded other

non-residential building types and did not consider environmental variables. In contrast, this review focuses on (1) field studies only to ensure real-world relevance, (2) all non-residential building types, including offices, schools, and healthcare facilities, and (3) quantitative modelling approaches, emphasising how environmental, personal, and contextual variables are associated with subjective IAQ assessments.

**Table 1.** Overview of existing literature reviews.

| Author, Year and Ref        | Subject Coverage                                                                                                                                                                                                                                   | Investigated Effect(s)                                                    |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| Seppänen et al. (1999) [40] | Review on the effects of CO <sub>2</sub> concentrations on human health and perception<br><i>Non-residential buildings</i>                                                                                                                         | Ventilation rate, CO <sub>2</sub> concentration                           |
| Wargocki et al. (2002) [35] | Review on the effects of ventilation on occupants' health, comfort and productivity<br><i>Non-industrial buildings (field studies and experiments)</i>                                                                                             | Ventilation rate                                                          |
| Toftum (2002) [32]          | Review on human perception of the indoor environment<br><i>Residential and non-residential</i>                                                                                                                                                     | Temperature, relative humidity, ventilation rate, noise level             |
| Wargocki (2004) [33]        | Review on data to quantify sensory pollution loads<br><i>Non-industrial buildings</i>                                                                                                                                                              | Sensory pollution loads                                                   |
| Clausen (2004) [36]         | Review on the effects of ventilation filters on occupants' perceived air quality, sick building syndrome symptoms and productivity<br><i>Experiments and field studies performed by the International Centre for Indoor Environment and Energy</i> | Sensory pollution loads, ventilation filters                              |
| Sekhar (2016) [37]          | Review on thermal comfort issues and perceived air quality in air-conditioned buildings<br><i>Offices and educational buildings (field studies and experiments) in hot and humid climates</i>                                                      | Indoor air humidity                                                       |
| Wolkoff (2018) [34]         | Review on the effects of indoor air humidity on occupants' PAQ, health and work performance<br><i>Residential and non-residential buildings</i>                                                                                                    | Indoor air humidity                                                       |
| Pei et al. (2024) [31]      | Review on associations between objective IAQ variables and subjective IAQ<br><i>Residential buildings in China</i>                                                                                                                                 | Temperature, relative humidity, VOC, ventilation rate, particulate matter |

Table 1. Cont.

| Author, Year and Ref        | Subject Coverage                                                                                                                                  | Investigated Effect(s)                 |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| Torriani et al. (2024) [39] | Review on the assessment methods for subjective IAQ in office settings<br><i>Office environments (field study and experiments)</i>                | None                                   |
| Torriani et al. (2025) [38] | Review on building- and occupant-related factors on subjective IAQ in office settings<br><i>Office environments (field study and experiments)</i> | Building- and occupant-related factors |

### 1.3. Objectives of the Current Review

The objective of the current review is to systematically summarise field studies that investigate statistical or predictive models describing relationships between environmental, personal, and contextual variables and subjective IAQ assessment in non-residential buildings. Studies conducted in residential buildings, climate chambers, or involving personal comfort systems are excluded from the scope, as this review is focused on real-world, non-residential environments such as offices, schools, and healthcare facilities. This review specifically focuses on identifying which variables influence subjective IAQ and how these relationships are quantified using assessment scales and statistical or predictive modelling approaches applied in field studies. By synthesising existing research, this paper aims to provide a comprehensive overview of current methodologies and findings, and to identify key research gaps in the prediction and understanding of subjective IAQ assessments in non-residential settings. Based on these insights, this review offers recommendations to guide future field studies. Additionally, this review supports the integration of IAQ preferences into OCC systems by examining existing methodologies for predicting subjective IAQ and by providing an overview of confounding variables that influence occupants' subjective IAQ assessments. This review does not aim to compare methods of assessing objective and subjective IAQ.

## 2. Methods

The review methodology follows the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [41] (shown in Supplementary Materials) and the *JBIM Manual for Evidence Synthesis* [42]. The following steps were performed for the literature selection process: (1) delimiting the research scope, (2) designing a logic grid, (3) determining databases and selection criteria, (4) database search, (5) title and abstract screening, and (6) full paper screening.

### 2.1. Scope Delimiting and Logic Grid

The initial step in the review process comprised delimiting the research scope. Table 2 summarises the scope of the literature review. Based on the scope of the review, a logic grid was designed for the literature search. Table 3 represents the logic grid, which comprises keywords representing the scope of the review. Each column consisted of synonyms for the same keyword. Each column represented a different keyword, i.e., subjective IAQ, modelling, and non-residential.

**Table 2.** Scope of the current literature review.

| Focus of the Review                                                                    | Excluded from the Review                                                                                                                                                                                                                                     |
|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Subjective IAQ assessment                                                              | Studies focusing only on other domains of comfort (thermal, visual or acoustic comfort), studies investigating the impact of subjective IAQ assessment on occupants' wellbeing (e.g., performance, work engagement), or studies investigating objective IAQ. |
| Non-residential buildings                                                              | Residential buildings as well as studies focusing on subjective IAQ assessment in outdoor environments and vehicles                                                                                                                                          |
| Predictive models or statistical associations determined for subjective IAQ assessment | Studies that only evaluate or present descriptive statistics of subjective IAQ assessment                                                                                                                                                                    |
| Field measurement data                                                                 | Studies describing experimental data collecting in, e.g., climate chambers                                                                                                                                                                                   |
| Buildings naturally ventilated or ventilated by a total volume system                  | Studies focusing on personalised systems                                                                                                                                                                                                                     |

**Table 3.** Logic grid with keywords.

| Subjective IAQ                                                                                                                                                                                                                                                                                                                                   | Modelling                                                                | Non-Residential                                                                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| "subjective IAQ" OR<br>"subjective indoor air quality" OR<br>"perceived IAQ" OR<br>"perceived indoor air quality" OR<br>"perceived air quality" OR<br>"IAQ *satisfaction" OR<br>"indoor air quality *satisfaction" OR<br>"air quality *satisfaction" OR<br>"IAQ perception" OR<br>"indoor air quality perception" OR<br>"air quality perception" | model* OR<br>regression OR<br>relation* OR<br>association* OR<br>impact* | "Non-residential" OR<br>office* OR<br>school* OR<br>classroom* OR<br>commercial |

The symbol "\*" is used as a truncation symbol during the literature search in order to expand the search results by including different spellings or grammatical forms of a search term.

## 2.2. Determining the Databases and Selection Criteria

The literature search was performed in two databases, i.e., *Web of Science* and *Scopus*. The logic grid (Table 3) was translated into a search query in each database. Keywords in the same column were separated by the boolean operator "OR", leading to an increasing number of search results. On the contrary, each column was separated by the boolean operator "AND", which decreased the number of search results. Furthermore, modifiers (i.e., asterisk and quotation marks) were used to consider variations on keywords (e.g., satisfaction and dissatisfaction, relation and relationship, ...) and to keep multiple keywords together.

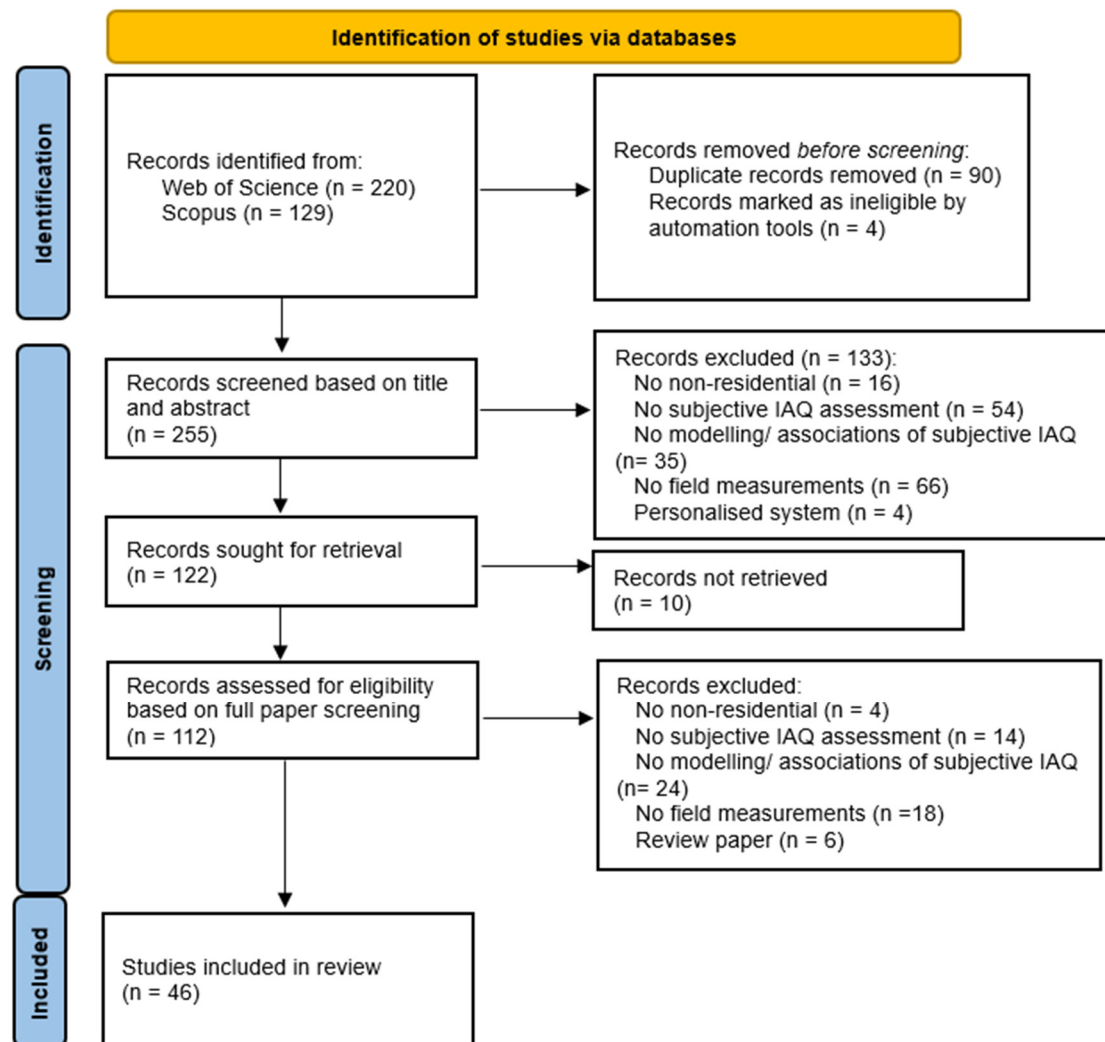
Only a language criterion was active for the database search, leading to only English-written papers being selected. No criterion on publication date was active, and both journal as well as peer-reviewed conference papers were eligible for selection. After the database search, the selected records were screened twice: firstly, a title and abstract screening, and secondly, a full paper screening. In both cases, the screening was performed based on the exclusion criteria derived from Table 2.

## 2.3. Database Search and Paper Screening

The initial search using the logic grid was performed in October 2025 and resulted in 215 papers in the *Web of Science* database and 126 papers in the *Scopus* database, respectively.



After duplicate removal, the title and abstract of 248 papers were screened (see Figure 1). After the title and abstract screening, the remaining 121 papers were sought for retrieval, of which 10 papers could not be accessed. Finally, 111 papers were fully screened (see Figure 1). The full paper screening led to 46 papers eligible for the current review, comprising 45 journal papers and 1 conference paper. An overview of the screening process is given in Figure 1, using the flowchart from the PRISMA guidelines.



**Figure 1.** Paper identification and screening flowchart based on PRISMA guidelines.

The 46 selected studies showed high heterogeneity in assessment scales and statistical techniques. Furthermore, some studies did not properly report the effect sizes of the statistical associations that were determined. Consequently, the current selection of papers is not suitable as a basis to perform a meta-analysis on the defined relationship between IEQ, personal, and contextual variables with subjective IAQ. Therefore, the current literature review focuses solely on reporting the current state-of-the-art on established associations and models for subjective IAQ assessment in non-residential field studies. As this review does not include a meta-analysis, no formal risk-of-bias and methodological quality assessment was conducted.

### 3. Results

Table 4 summarises the 46 papers studied in this review paper, ordered by their publication date. Table 4 shows information on the study context (i.e., geographical location,

room type, ventilation system, and type of study). Furthermore, the subjective IAQ assessment scale(s) that were used in the reviewed studies are presented. Moreover, the dataset and sample size, on which the analysis in the reviewed papers were performed, are given. Finally, the statistically significant IEQ-related, personal, and contextual variables are summarised together with the direction of the association, i.e., negative (-) or positive (+).



**Table 4.** Summarising table of 46 reviewed studies ordered by publication date.

| Authors, Year and Ref             | Study Location | Study Type <sup>(1)</sup>  | Room Type-Ventilation System <sup>(2)</sup>                | Subjective IAQ Assessment Scale                             | Dataset Size (No. Participants) | Statistical Method/Modelling Technique <sup>(3)</sup> | Associations with IEQ-Related Variables <sup>(4)</sup>                                                                                                                                                                    | Associations with Personal Variables                                                     | Associations with Contextual Variables                                                                                                            |
|-----------------------------------|----------------|----------------------------|------------------------------------------------------------|-------------------------------------------------------------|---------------------------------|-------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Hedge et al. (1996) [43]          | USA            | Retrospective              | Office<br><i>Balanced MV: VAV + CAV with recirculation</i> | IAQ complaint frequency (4p)                                | 4372                            | LogR                                                  | T (-), RH (-), HCHO (-), CO <sub>2</sub> (-)                                                                                                                                                                              | /                                                                                        | /                                                                                                                                                 |
| Smedje et al. (1997) [44]         | Sweden         | Retrospective              | Classroom<br><i>Balanced MV + NV</i>                       | Air quality acceptability (4p)                              | 1410                            | LogR, LinR                                            | TVOC (-), PM (-), mould (-), bacteria (-)                                                                                                                                                                                 | Age (+), allergy (-), psychosocial school climate (+), smoke exposure (+), pet owner (+) | Displacement ventilation (+), School building age (+)                                                                                             |
| Haghighat and Donnini (1999) [45] | n/a            | RHRN                       | Office<br><i>MV</i>                                        | /                                                           | 877                             | Corr (Pearson)                                        | /                                                                                                                                                                                                                         | Personal control (+)                                                                     | /                                                                                                                                                 |
| Mui and Wong (2007) [46]          | China          | RHRN                       | Office<br><i>MV: CAV</i>                                   | Air quality acceptability (5p, binary)                      | 396                             | LogR                                                  | CO <sub>2</sub> (-)                                                                                                                                                                                                       | /                                                                                        | /                                                                                                                                                 |
| Kostiainen et al. (2008) [47]     | n/a            | Longitudinal               | Office<br><i>MV: CAV</i>                                   | IAQ satisfaction (3p)                                       | 639 (109)                       | Hierarchical Bayesian probability model               | T (-)                                                                                                                                                                                                                     | Allergy (-), Mental wellbeing (+)                                                        | /                                                                                                                                                 |
| Bakke et al. (2008) [48]          | Norway         | Retrospective              | Office<br><i>Balanced MV</i>                               | IAQ complaint frequency (3p)                                | 173                             | LogR                                                  | /                                                                                                                                                                                                                         | Biomarker [MCA] (+)                                                                      | /                                                                                                                                                 |
| Norbäck et al. (2011) [49]        | Sweden         | Intervention study         | Classroom<br><i>Balanced MV</i>                            | Air quality acceptability (6p)<br>IAQ perception odour (6p) | 183 (61)                        | Mann–Whitney U test                                   | /                                                                                                                                                                                                                         | /                                                                                        | Displacement ventilation (+)                                                                                                                      |
| Wang et al. (2015) [50]           | Sweden         | Retrospective Longitudinal | Classroom<br><i>n/a</i>                                    | Air quality acceptability (4p)                              | 3330 (2034)                     | Mann–Whitney U test, Linear ME                        | CO <sub>2</sub> (-), illuminance (+),<br>T (-), CO <sub>2</sub> (-), RH (-), NO <sub>2</sub> (-), O <sub>3</sub> (-), T4CE (-), benzene (-), limonene (-), pinene (-), HCHO (-), PM (-), TVOC (-), microbial exposure (-) | Age (-), Allergy (-)                                                                     | ACH (+)                                                                                                                                           |
| Chatzidiakou et al. (2015) [51]   | UK             | Retrospective              | Classroom<br><i>n/a</i>                                    | IAQ satisfaction (7p)                                       | 752 (376)                       | Ordinal ME                                            | T (-), CO <sub>2</sub> (-), RH (-), NO <sub>2</sub> (-), O <sub>3</sub> (-), T4CE (-), benzene (-), limonene (-), pinene (-), HCHO (-), PM (-), TVOC (-), microbial exposure (-)                                          | Age (+), Stress (-), Satisfaction psychosocial climate school (+), Exposure to smoke (+) | ACH (+), Distance from traffic (+), cleaning end of day (+), visible mould (-), occupancy density (-), openable window area (+), curtain area (-) |
| Markowicz and Larsson (2015) [52] | n/a            | Intervention               | Classroom<br><i>n/a</i>                                    | Odour intensity (continuous)                                | 22                              | Mann–Whitney U test                                   | /                                                                                                                                                                                                                         | /                                                                                        | Surface emission trap device (+)                                                                                                                  |
| Thatcher and Milner (2016) [53]   | n/a            | Intervention               | Office<br><i>MV: CAV</i>                                   | IAQ complaint frequency (4p)                                | 560 (280)                       | Paired <i>t</i> -test                                 | /                                                                                                                                                                                                                         | /                                                                                        | Green building (+)                                                                                                                                |
| Cablé et al. (2016) [54]          | Norway         | Intervention               | Classroom<br><i>MV: DCV</i>                                | Odour perception (continuous)                               | 57 (19)                         | Linear ME                                             | /                                                                                                                                                                                                                         | /                                                                                        | /                                                                                                                                                 |
| Finell et al. (2017) [55]         | Finland        | Retrospective              | Classroom<br><i>n/a</i>                                    | IAQ complaint frequency (4p)                                | 26,946                          | Linear ME                                             | /                                                                                                                                                                                                                         | Girl (-), Perceived student- teacher relationship (+), stress (-)                        | /                                                                                                                                                 |

Table 4. Cont.

| Authors, Year and Ref                | Study Location | Study Type <sup>(1)</sup>  | Room Type-Ventilation System <sup>(2)</sup>                 | Subjective IAQ Assessment Scale                        | Dataset Size (No. Participants) | Statistical Method/Modelling Technique <sup>(3)</sup>  | Associations with IEQ-Related Variables <sup>(4)</sup> | Associations with Personal Variables                          | Associations with Contextual Variables                                                             |
|--------------------------------------|----------------|----------------------------|-------------------------------------------------------------|--------------------------------------------------------|---------------------------------|--------------------------------------------------------|--------------------------------------------------------|---------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Kang et al. (2017) [56]              | China          | Retrospective              | Office<br><i>n/a</i>                                        | IAQ satisfaction (5p)                                  | 231                             | Mann–Whitney U test, Kruskal–Wallis test               | /                                                      | Male (+), Birthplace (North China +),                         | Sitting near door or window (+)                                                                    |
| Cheung et al. (2017) [57]            | Singapore      | Longitudinal               | Office<br><i>n/a</i>                                        | Air quality acceptability (continuous)                 | 545 (15)                        | LinR, GP                                               | /                                                      | /                                                             | /                                                                                                  |
| Finell et al. (2018) [58]            | Finland        | Retrospective Longitudinal | Classroom<br><i>n/a</i>                                     | IAQ complaints frequency (4p)                          | 1972 (986)                      | LinR, LCM                                              | /                                                      | Mental wellbeings (+), Perceived teacher-student problems (-) | /                                                                                                  |
| Järvi et al. (2018) [59]             | Finland        | Longitudinal               | Classroom<br><i>Balanced MV</i>                             | IAQ complaints (binary)                                | 7041 (1373)                     | Corr (Pearson’s Chi-squared test, Fisher’s exact test) | /                                                      | /                                                             | /                                                                                                  |
| Vornanen-Winqvist et al. (2018) [60] | Finland        | Intervention               | Classroom<br><i>Balanced MVHR</i>                           | IAQ complaints frequency (3p)                          | 75                              | Corr (Fisher’s exact test)                             | /                                                      | /                                                             | /                                                                                                  |
| Tahtinen et al. (2018) [61]          | Finland        | Retrospective              | Workplace hospital<br><i>n/a</i>                            | IAQ complaints frequency (3p)                          | 2669                            | Mann–Whitney U test                                    | /                                                      | /                                                             | /                                                                                                  |
| Park et al. (2019) [62]              | USA            | RHRN                       | Office<br><i>n/a</i>                                        | IAQ satisfaction (7p)                                  | 1051                            | T-test, Corr (Pearson’s Chi-squared test)              | CO <sub>2</sub> (-), PM (-)                            | /                                                             | Operable window (+), Dedicated exhausts (+), Return air diffuser density (-), Partition height (+) |
| Liu et al. (2019) [63]               | China          | RHRN                       | Classroom<br><i>NV</i>                                      | Odour intensity (6p)<br>Air quality acceptability (4p) | 992                             | LinR                                                   | /                                                      | /                                                             | Occupancy density (-)                                                                              |
| Lind et al. (2019) [64]              | Norway         | Intervention               | Office<br><i>Balanced MV; DCV</i>                           | Air quality acceptability (continuous)                 | /                               | Friedman’s test                                        | RH (-)                                                 | /                                                             | /                                                                                                  |
| Thach et al. (2020) [65]             | Singapore      | Retrospective              | Workplace (office, control and workshop room)<br><i>n/a</i> | IAQ satisfaction (7p)                                  | 464                             | Kruskall–Wallis test                                   | /                                                      | Stress (-)                                                    | /                                                                                                  |
| Ma et al. (2020) [66]                | China          | Longitudinal               | Classroom<br><i>NV</i>                                      | Air freshness (4p)<br>IAQ satisfaction (5p)            | 853 (141)                       | LinR                                                   | /                                                      | /                                                             | /                                                                                                  |
| Korsavi et al. (2021) [67]           | UK             | Longitudinal               | Classroom<br><i>NV</i>                                      | Air freshness (5p, binary)                             | 1390 (805)                      | LinR, Corr (Spearman correlation)                      | T (-), CO <sub>2</sub> (-)                             | /                                                             | /                                                                                                  |

Table 4. Cont.

| Authors, Year and Ref          | Study Location  | Study Type <sup>(1)</sup>  | Room Type-Ventilation System <sup>(2)</sup>    | Subjective IAQ Assessment Scale                                              | Dataset Size (No. Participants) | Statistical Method/Modelling Technique <sup>(3)</sup> | Associations with IEQ-Related Variables <sup>(4)</sup> | Associations with Personal Variables | Associations with Contextual Variables |
|--------------------------------|-----------------|----------------------------|------------------------------------------------|------------------------------------------------------------------------------|---------------------------------|-------------------------------------------------------|--------------------------------------------------------|--------------------------------------|----------------------------------------|
| Finell et al. (2021) [68]      | Finland         | Retrospective              | Classroom<br><i>n/a</i>                        | IAQ complaint frequency (3p)                                                 | 25,101                          | Linear ME                                             | /                                                      | /                                    | Visible mould (-)                      |
| Licina and Langer (2021) [69]  | The Netherlands | Intervention               | Office<br><i>MV</i>                            | IAQ satisfaction (7p)                                                        | 786 (250)                       | Corr (Spearman), Mann–Whitney U test                  | /                                                      | /                                    | WELL certified (+)                     |
| de Oliveira et al. (2021) [70] | Brazil          | RHRN                       | Office<br><i>MV, NV</i>                        | IAQ satisfaction (7p)                                                        | 7564                            | LinR                                                  | T (-)                                                  | /                                    | /                                      |
| Jung and Awad (2021) [71]      | UAE             | Intervention               | Classroom<br><i>n/a</i>                        | Air quality acceptability (continuous)<br>Air freshness (continuous)         | /                               | <i>t</i> -test                                        | /                                                      | /                                    | Presence of plants (+)                 |
| Yang et al. (2021) [20]        | Norway          | Intervention               | Classroom<br><i>Balanced MV: DCV</i>           | Air quality acceptability (continuous)<br>Odour intensity (6p)               | 73                              | paired <i>t</i> -test, LinR                           | Enthalpy (-)                                           | /                                    | /                                      |
| Xie et al. (2021) [72]         | China           | RHRN                       | Badminton hall<br><i>NV</i>                    | Air quality acceptability (4p)<br>Odour intensity (4p)<br>Air freshness (4p) | 185                             | Corr (Pearson)                                        | /                                                      | /                                    | /                                      |
| Kong et al. (2022) [73]        | South Korea     | Longitudinal, Intervention | Office<br><i>n/a</i>                           | Air quality acceptability (3p)                                               | 1280 (10)                       | GBM                                                   | T (/), RH (/), TVOC (/), CO <sub>2</sub> (/), PM (/)   | /                                    | IEQ info provided (/)                  |
| Orru et al. (2022) [74]        | Sweden          | Retrospective              | Office<br><i>Balanced MVHR</i>                 | Air quality acceptability (5p)                                               | 260                             | SEM                                                   | /                                                      | /                                    | /                                      |
| Choe et al. (2022) [75]        | South Korea     | Longitudinal               | Classroom<br><i>NV</i>                         | IAQ satisfaction (7p)                                                        | /                               | Corr                                                  | CO <sub>2</sub> (-)                                    | /                                    | /                                      |
| Shim et al. (2023) [76]        | South Korea     | Retrospective              | Store<br><i>Balanced MV</i>                    | IAQ complaints frequency (5p)                                                | 77                              | Corr (Spearman)                                       | RH (-), HCHO (-), benzene (-)                          | /                                    | /                                      |
| Torriani et al. (2023) [77]    | Italy           | RHRN                       | Classroom<br><i>NV, MV</i>                     | Air quality acceptability (continuous)                                       | 585                             | LinR                                                  | T (-)                                                  | Personal control (+)                 | /                                      |
| Kuurola et al. (2023) [78]     | Finland         | Intervention               | Classroom and Playroom<br><i>Balanced MVHR</i> | IAQ complaints frequency (3p)                                                | 298 (189)                       | Linear ME                                             | T (-), CO <sub>2</sub> (+)                             | /                                    | /                                      |
| Parhizkar et al. (2023) [79]   | Thailand        | RHRN                       | Office<br><i>NV, MV: VAV</i>                   | IAQ satisfaction (5p)                                                        | 579                             | LogR                                                  | CO <sub>2</sub> (-)                                    | /                                    | /                                      |
| Wu and Wagner (2024) [80]      | China           | Longitudinal               | Classroom<br><i>NV</i>                         | Air quality acceptability (continuous)                                       | 2349 (250)                      | LinR, OrdR                                            | T (+/-)                                                | /                                    | /                                      |

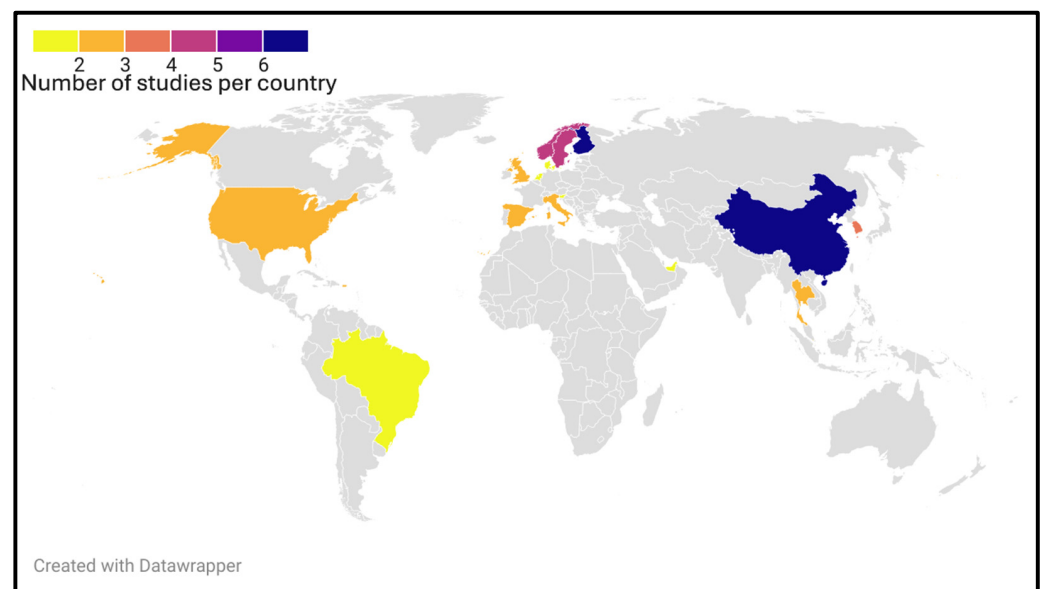
Table 4. Cont.

| Authors, Year and Ref               | Study Location | Study Type <sup>(1)</sup> | Room Type-Ventilation System <sup>(2)</sup> | Subjective IAQ Assessment Scale                                | Dataset Size (No. Participants) | Statistical Method/Modelling Technique <sup>(3)</sup>  | Associations with IEQ-Related Variables <sup>(4)</sup> | Associations with Personal Variables | Associations with Contextual Variables                                                                                               |
|-------------------------------------|----------------|---------------------------|---------------------------------------------|----------------------------------------------------------------|---------------------------------|--------------------------------------------------------|--------------------------------------------------------|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Li et al. (2024) [81]               | China          | Longitudinal              | Office MV                                   | Air quality acceptability (continuous)<br>Odour intensity (6p) | 469 (23)                        | Mann–Whitney U test                                    | /                                                      | /                                    | /                                                                                                                                    |
| Surawattanasakul et al. (2024) [82] | Thailand       | Retrospective             | Office n/a                                  | IAQ complaints frequency (5p)                                  | 261                             | LinR                                                   | CO <sub>2</sub> (-), HCHO (-)                          | /                                    | /                                                                                                                                    |
| Sørensen and Kristensen (2024) [83] | Denmark        | Retrospective             | Classroom NV + MV                           | Air quality acceptability (4p)                                 | 75                              | Corr (Spearman)                                        | TVOC (-)                                               | /                                    | /                                                                                                                                    |
| Galičič et al. (2024) [84]          | Slovenia       | Retrospective             | Classroom NV + MV                           | Air quality acceptability (5p)                                 | 355                             | Corr (Pearson’s Chi-squared test, Fishers’ exact test) | /                                                      | /                                    | City center location (-), proximity road (-), proximity wood burning stoves (-), wet cleaning (+), manual airing during cleaning (+) |
| Alonso et al. (2025) [85]           | Spain          | Longitudinal              | Classroom NV                                | Odour intensity (4p)<br>Air quality acceptability (5p)         | 1056 (ns)                       | LinR                                                   | T (-)                                                  | /                                    | /                                                                                                                                    |
| Miao et al. (2025) [86]             | Spain          | Longitudinal              | Classroom NV                                | Air quality acceptability (7p)                                 | 421 (30)                        | Corr (Spearman)                                        | CO <sub>2</sub> (-)                                    | /                                    | Occupancy ratio (+), window opening area (+)                                                                                         |
| Gnecco et al. (2025) [87]           | Italy          | Longitudinal              | Office MV                                   | Air quality acceptability (5p)                                 | 993 (ns)                        | GNB                                                    | CO <sub>2</sub> (-), T (-)                             | /                                    | /                                                                                                                                    |

(1) RHRN: right-here-right-now survey campaign. (2) NV: natural ventilation, MV: mechanical ventilation, VAV: variable air volume, CAV: constant air volume, MVHR: mechanical ventilation with heat recovery, DCV: demand-controlled ventilation. (3) LogR: logistic regression, LinR: linear regression, Corr: correlation analysis, OrdR: ordinal regression, ME: mixed-effects model, GP: Gaussian process, LCM: latent change model, GBM: gradient boosting machine, SEM: structural equation model, GNB: Gaussian Naïve Bayes probability model. (4) T: indoor temperature [°C], RH: relative humidity [%], HCHO: formaldehyde [µg/m<sup>3</sup>], CO<sub>2</sub>: carbon dioxide [ppm], O<sub>3</sub>: ozone [µg/m<sup>3</sup>], PM: particulate matter [µg/m<sup>3</sup>], TVOCs: total volatile organic compounds [µg/m<sup>3</sup>], T4CE: tetrachloroethylene [µg/m<sup>3</sup>].

### 3.1. Study Context

Figure 2 shows the geographical locations of the studies. Most of them were performed in China (7) and Finland (7), followed by Norway (4), Sweden (4), South Korea (3), the USA (2), the UK (2), Thailand (2), Italy (2), Spain (2), and Singapore (2). UAE, The Netherlands, Slovenia, Denmark, and Brazil were each mentioned in only one study. A total of 4 out of the 46 studies did not specify the location of the investigated building [45,47,52,53]. The widespread geographic locations also led to multiple climate zones being represented in the studies, ranging from tropical [57,65,71,79,82] and subtropical [46,63,70,72,73,75,77,80,81] to cool [51,67,69,83] to cold climates [20,44,48–50,55,58–61,64,68,74,78] following the ASHRAE climate classification [88]. There is an underrepresentation of studies in climate zones that are warm and dry or temperate. The climate conditions, i.e., outdoor temperatures and humidity, could influence the associations determined between variables and subjective IAQ. Therefore, results obtained in a specific climate zone cannot be generalised to another climate zone.



**Figure 2.** Map chart with the distribution of the studies' geographical location. (Software version 2025).

As shown in Table 4, most of the room types studied were offices (18 studies) and classrooms (24 studies). Other room types, such as a badminton hall [72], a hospital workplace [61], and a store [76], have only been topics in one of the reviewed studies. Thach et al. [65] and Kuurola et al. [78] investigated subjective IAQ in multiple room types. The first study investigated workers' subjective IAQ in three different workplaces, i.e., office, control, and workshop rooms. The latter studied employees' and children's subjective IAQ in classrooms of primary schools and in playrooms of daycare centres.

The ventilation system type also varied among the reviewed papers. Both naturally ventilated (14 studies) and mechanically ventilated spaces (23 studies) were investigated. In the case of mechanically ventilated spaces, the authors did not always describe the characteristics of the ventilation system. Ten studies clarified that the mechanical ventilation system was a balanced system, of which three specified the presence of a heating recovery. Only one study [20] indicated that the investigated room was equipped with a demand-controlled ventilation system. Fourteen studies did not detail the characteristics of the ventilation system present in the studied room(s).

### 3.2. Study and Survey Type

Four study types can be distinguished from the reviewed papers in Table 4, i.e., (1) retrospective (16 studies), (2) right-here-right-now (RHRN) (8 studies), (3) intervention (11 studies), and (4) longitudinal (14 studies). Each study type varies in the survey type used for assessing participants' subjective IAQ, the study duration, and the study goal. A retrospective and RHRN study differed in the recall period of the surveys used to evaluate participants' perception or satisfaction with the IAQ. A retrospective survey evaluated participants' subjective IAQ in relation to a certain period prior to the moment of assessment. This recall period ranged from one week [51] or one month [43,58,65,76] to multiple months [48,61] or one year [68]. Some studies did not specify a recall period [56,74]. On the other hand, an RHRN survey assessed participants' perceptions at the moment of completing the survey. The advantage of a retrospective survey is that it can provide a general assessment of the participants' perceptions for a long period, while an RHRN survey only gives a result for a specific moment in time. However, an RHRN survey is more suitable when the aim is to investigate the associations between IEQ conditions and contextual variables that vary in time (e.g., window opening) with subjective IAQ, given that the RHRN survey is combined with simultaneous monitoring of the IEQ and contextual conditions. Associating IEQ or contextual conditions with the results of a retrospective survey is more challenging since it is unclear to which conditions the survey results relate. If a retrospective study investigated the associations between subjective IAQ and IEQ-related variables, average IEQ conditions over the assessed period were used. Furthermore, retrospective surveys are also prone to a recall bias [89], especially when a large recall period is applied. The RHRN and retrospective study type referred to studies in which this type of survey was completed only once by each participant.

On the contrary, a study is categorised as longitudinal when the participants' subjective IAQ was assessed multiple times during the study duration (i.e., multiple times per day [57,66,73,81], each day [47], multiple times per week [59], each half month [80], each month [75], multiple times per year [67], and once each two years [50,58]). The type of survey can be both retrospective and RHRN. A longitudinal study makes it possible to investigate the impact of time-dependent variables (i.e., variables that change over time) on subjective IAQ.

The last type of studies were intervention studies. Intervention studies investigate the impact of a certain intervention (i.e., changing IAQ management strategy [49,52,54,60], changing building characteristics [53,69], varying IEQ conditions [20,64], presence of indoor plants [71], and informing occupants of current IEQ conditions [73]) on the subjective IAQ assessment. Intervention studies often consist of a control and test group or perform a pre- and post-test to quantify the influence of the intervention.

### 3.3. Dataset and Sample Size

Table 4 shows that the dataset sizes varied significantly among the different studies, ranging from as small as 22 datapoints [52] to as large as 26,946 datapoints [55]. Furthermore, the number of participants varied from 10 occupants [73] to 26,946 [55]. A median sample size of 585 datapoints and 261 participants was applied. Three of the reviewed studies did not specify the sample size [64,71,75]. Many studies only evaluate participants' subjective IAQ once during the study period (22 studies). Sixteen of the reviewed studies evaluate participants' subjective IAQ multiple times during the study duration. The ratio between the number of datapoints and participants ranged from 1.6 datapoints/person [78] to 128 datapoints/person [73]. The number of datapoints per person was related to the aim of the study. Kuurola et al. [78] aimed to assess the impact of two ventilation operation modes on occupants' subjective IAQ. Therefore, participants were only required to assess



their subjective IAQ no more than twice during the study. On the contrary, Kong et al. [73] intended to investigate the effect of providing information about the objective IAQ on occupants' subjective IAQ assessment. Furthermore, the authors wanted to examine the continuity of this effect over time; consequently, participants were exposed to 128 assessment cycles throughout a 16-day measurement period.

In the reviewed papers, the discussion of the required sample size was often limited to stating a need for a larger sample size to generalise the statistical results obtained in the study. Only one of the reviewed studies elaborated on a power analysis calculation to determine the minimal sample size to obtain sufficient statistical power [87]. Furthermore, two studies described a required sample size without elaborating on the reason for this number. Haghighat and Donnini [45] set 40 occupants per monitored workstation as a target sample size for their study. Licina and Langer [69] described that 50 occupants per measured building were required as the minimum sample size.

### 3.4. Subjective IAQ Assessment Scales

Table 5 gives a detailed description of the subjective IAQ assessment questions and scales. The evaluation method for subjective IAQ is divided into 5 categories, i.e., (1) IAQ complaint frequency, (2) air quality acceptability, (3) IAQ satisfaction, (4) odour intensity, and (5) air freshness. The first evaluates the IAQ conditions based on how frequently occupants perceive a problem with the IAQ, such as unpleasant odours, stale, humid, or dry or dusty air. The description of the scale points used to assess the frequency of perceiving IAQ problems can be specific (e.g., "1–3 times/month", "yes weekly", or "1–3 times/week") or more general (e.g., "sometimes", "a little", or "a lot"). Four studies [48,60,61,78] applied the same three-point scale to evaluate how frequently occupants are bothered by stuffy air and unpleasant odours. The scale originated from the MM40 questionnaire [90], which is a retrospective survey to assess occupants' perception in their school or workplace over three months prior to the moment of completing the survey. The MM40 questionnaire had been mostly applied in Nordic countries in Europe [90]. Hedge et al. [43] and Thatcher and Milner [53] both used the same four-point scale ("never", "1–3 times/month", "1–3 times/week", or "every day") to assess the frequency of IAQ complaints. However, the survey question related to the scale was not specified in either of the studies. The other studies measured the IAQ complaints frequency on a binary [59], three-point [68], four-point [55,58], or five-point scale [76,82].

A second method to measure occupants' subjective IAQ was to evaluate their acceptability of the air quality. In most studies, air quality acceptability was evaluated on a non-divided continuous scale, mostly ranging from "clearly unacceptable" to "clearly acceptable" [64,71,77,81]. Three of the reviewed studies assessed the air quality acceptability on a continuous scale that is divided into two equal parts [16,31,41]. The missing mid-point in the divided scale urged occupants to choose between "just unacceptable" and "just acceptable" instead of a mid-point indicating neutrality. Five studies [44,50,63,72,83] applied a four-point scale without a mid-point, similar to the divided continuous acceptability scale. Mui and Wong [46] assessed the indirect and direct IAQ acceptability on a five-point and binary scale, respectively. Kong et al. [73] and Orru et al. [74] measured the air quality acceptability on a three- and five-point Likert scale, respectively, ranging from "(very) bad" to "(very) good" and with "acceptable" as a mid-point. One study assessed the air quality acceptability on a six-point scale, of which only the ends were labelled, i.e., "Extremely poor" and "Extremely good" [49]. Finally, a seven-point scale was used in the study of Miao et al. [86] to evaluate occupants' IAQ acceptability in a classroom.

**Table 5.** Overview of the assessment scales used for subjective IAQ assessment and questions asked prior to the assessment. The table is divided per scale category: (1) IAQ complaint frequency, (2) Air quality acceptability, (3) IAQ satisfaction, (4) odour intensity, and (5) air freshness.

| Category                  | Question                                                                                                                                                                                                                                            | Scale                                                                                             | Ref           |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|---------------|
| IAQ complaint frequency   | not specified                                                                                                                                                                                                                                       | 4p: “never”, “1–3 times/month”, “1–3 times/week”, or “every day”                                  | [43,53]       |
|                           | “Have you been bothered during the last three months by stuffy ‘bad’ air/unpleasant odours at your work place?”                                                                                                                                     | 3p: “yes weekly”, “yes sometimes”, or “not at all”                                                | [48,60,61,78] |
|                           | “In your school, do the following conditions disturb your school work? Insufficient ventilation or bad indoor air”                                                                                                                                  | 4p: “not at all”, “rather little”, “rather much”, or “very much”                                  | [55]          |
|                           | “In your school, have the following conditions disturbed you schoolwork during last month? (a) stuffy (bad) air; (b) dust, dirt; (c) unpleasant odour”                                                                                              | 4p: “not at all”, “a little”, “quite a lot”, or “very much”                                       | [58]          |
|                           | (1) “Did the poor indoor air quality bother you during the last hour?” with follow up question (2) “What bothered?” (stuffy air, fragrance, smell, . . .)                                                                                           | (1) binary: “yes” or “no”<br>(2) binary: “little” or “much”                                       | [59]          |
|                           | “Have any of the following things bothered you at your school during this school year? (a) ‘Stuffy air (bad indoor air)’; (b) ‘unpleasant odour’”                                                                                                   | 3p: “not at all”, “some”, or “a lot”                                                              | [68]          |
|                           | “Have you experienced the following perception while working in your stores in the USD within the last month: (1) stuffy odour; (2) unpleasant odour; (3) pungent odour; (4) mouldy odour; (5) tobacco smoke odour; (6) humid air; or (7) dry air?” | 5p: “never”, “rarely”, “neutral”, “sometimes”, or “frequently”                                    | [76]          |
|                           | Participants were asked if they experienced discomfort (i.e., stuffy bad air and unpleasant odour) during the past three months                                                                                                                     | 5p: “3–5 days per week”, “1–2 days per week”, “2–3 times per month”, “once per month”, or “never” | [82]          |
| Air quality acceptability | “How do you perceive the quality of air inside the school?”                                                                                                                                                                                         | 4p: “very good”, “good”, “bad”, or “very bad”                                                     | [44]          |
|                           | not specified                                                                                                                                                                                                                                       | 5p: “Very bad”, “Bad”, “Neutral”, “Good”, or “Very good”                                          | [46]          |
|                           | “Is the indoor air quality being perceived in the office environment acceptable to you?”                                                                                                                                                            | Binary: “Acceptable” or “Not acceptable”                                                          | [46]          |
|                           | not specified                                                                                                                                                                                                                                       | 6p: “Extremely poor”, “. . .”, or “Extremely good”                                                | [49]          |
|                           | “On average, how do you perceive the indoor air quality (IAQ) in your classroom?”                                                                                                                                                                   | 4p: “very poor”, “fairly poor”, “fairly good”, or “very good”                                     | [50]          |

Table 5. Cont.

| Category                  | Question                                                              | Scale                                                                                                                                                                    | Ref  |
|---------------------------|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| Air quality acceptability | not specified                                                         | Continuous: 1–0.01 (–0.01)–(–1)<br>("Clearly acceptable" to "just acceptable" to "just unacceptable" to "clearly unacceptable")                                          | [57] |
|                           | "How do you perceive the indoor air quality?"                         | 4p: "Clearly unacceptable", "just unacceptable", "just acceptable", or "clearly acceptable"                                                                              | [63] |
|                           | "What is your perception of the air quality in this room?"            | Continuous: 1–5–10 ("Clearly unacceptable" to "just unacceptable/just acceptable" to "clearly acceptable")                                                               | [64] |
|                           | not specified                                                         | Continuous: 0–5 ("comfort" to "intolerance")                                                                                                                             | [71] |
|                           | "Mark your perception of the air quality"                             | Continuous: –1–0 0–1 ("Clearly unacceptable" to "just unacceptable" to "just acceptable" to "clearly acceptable")                                                        | [20] |
|                           | "How do you perceive the indoor air quality at the moment?"           | 4p: "clearly unacceptable", "just unacceptable", "just acceptable", or "clearly acceptable"                                                                              | [72] |
|                           | not specified                                                         | 3p: "bad", "acceptable", or "good"                                                                                                                                       | [73] |
|                           | "What do you think about the air quality at the work place?"          | 5p: "very good", "good", "acceptable", "bad", or "very bad"                                                                                                              | [74] |
|                           | "How do you judge the air quality in the room at the moment?"         | Continuous: –3–+3 ("Terrible" to "Excellent")                                                                                                                            | [77] |
|                           | not specified                                                         | Continuous: –2–(–1)–(–0.01) 0.01–1–2 ("Clearly unacceptable" to "Unacceptable" to "Slightly unacceptable" to "Slightly acceptable" to "Acceptable" to "Very acceptable") | [80] |
|                           | "At this moment, how do you assess the IAQ?"                          | Continuous: (–5)–(+5) ("Entirely unacceptable" to "Entirely acceptable")                                                                                                 | [81] |
|                           | "Perceived air quality"                                               | 4p: "Poor", "Less acceptable", "Acceptable", or "Good"                                                                                                                   | [83] |
|                           | Not specified                                                         | 5p: "Very good", "Good", "Not good, not bad", "Bad", or "Very bad"                                                                                                       | [84] |
|                           | "How would you rate the air quality in the classroom at this moment?" | 5p: "Acceptable", "Slightly improvable", "Improvable", "Very improvable", or "Unacceptable"                                                                              | [85] |

Table 5. Cont.

| Category                  | Question                                                                          | Scale                                                                                                                                                 | Ref           |
|---------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| Air quality acceptability | "How do you feel about the air quality in the classroom?"                         | 7p: "Terrible", "Bad", "Poor", "Neutral", "Fine", "Good", or "Excellent"                                                                              | [86]          |
|                           | Not specified                                                                     | 5p: "Comfortable", "Slightly uncomfortable", "Uncomfortable", "Very uncomfortable", or "Extremely uncomfortable"                                      | [87]          |
| IAQ satisfaction          | not specified                                                                     | 3p: "very satisfied", "neutral", or "very dissatisfied"                                                                                               | [47]          |
|                           | not specified                                                                     | 7p: "Very dissatisfied", "Dissatisfied", "Slightly/somewhat dissatisfied", "Neutral", "Slightly/somewhat satisfied", "Satisfied", or "Very satisfied" | [51,62,65,70] |
|                           | not specified                                                                     | 5p: "Very dissatisfied", "Dissatisfied", "Neutral", "Satisfied", or "Very satisfied"                                                                  | [56]          |
|                           | "How do you perceive the indoor air quality in the classroom?"                    | 5p: "Very satisfied", "Satisfied", "Neutral", "Dissatisfied", or "Very dissatisfied"                                                                  | [66]          |
|                           | "How satisfied are you with the air quality of your workspace?"                   | 7p: "Very dissatisfied", "Dissatisfied", "Somewhat dissatisfied", "Neutral", "Somewhat satisfied", "Satisfied", or "Very satisfied"                   | [69]          |
|                           | "In general, how satisfied are you with the overall air quality in the building?" | 5p: "Very dissatisfied", "slightly dissatisfied", "neither satisfied nor dissatisfied", "moderately satisfied", or "extremely satisfied"              | [79]          |
| Odour intensity           | not specified                                                                     | 6p: "No odour", "...", or "Extremely strong odour"                                                                                                    | [49]          |
|                           | not specified                                                                     | continuous: 0–80 ("no odour" to "extremely annoying odour")                                                                                           | [52]          |
|                           | "Is there any unpleasant smell"                                                   | Continuous: 0–1 ("very comfortable" to "very uncomfortable")                                                                                          | [54]          |
|                           | "How do you perceive the odour intensity?"                                        | 6p: "overpowering", "very strong odour", "strong odour", "moderate odour", "weak odour", or "no odour"                                                | [63]          |
|                           | "Mark your perception of the odour intensity"                                     | 6p: "no odour", "weak odour", "moderate odour", "strong odour", "very strong odour", or "overpowering odour"                                          | [20]          |

Table 5. Cont.

| Category        | Question                                                                               | Scale                                                                                                                       | Ref  |
|-----------------|----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|------|
| Odour intensity | "How do you perceive the odour intensity at the moment?"                               | 4p: "strong odour", "moderate odour", "weak odour", or "no odour"                                                           | [72] |
|                 | "The odour intensity"                                                                  | 6p: "not odorous", "slight odorous", "moderate odorous", "strong odorous", "very strong odorous", or "overwhelming odorous" | [81] |
|                 | "From your point of view, what do you think the classroom smells like at this moment?" | 4p: "It doesn't smell", "Slightly smelly", "Smelly", or "Very smelly"                                                       | [85] |
| Air freshness   | "How do you feel about the air freshness of your classroom at this moment?"            | 4p: "fresh", "stale", "stale with occasional odour", or "stale with odour"                                                  | [66] |
|                 | "How is the air in the classroom now?"                                                 | 5p: "very fresh", "fresh", "ok", "stuffy", or "very stuffy"                                                                 | [67] |
|                 | "I like the air to be . . . now"                                                       | Binary: "fresher" or "as it is"                                                                                             | [67] |
|                 | not specified                                                                          | Continuous: −3–+3 ("very stale" to "very fresh")                                                                            | [71] |
|                 | "How do you perceive the air freshness at the moment?"                                 | 4p: "fairly stale", "slightly stale", "slightly fresh", or "fairly fresh"                                                   | [72] |

Nine of the reviewed studies evaluated occupants' IAQ satisfaction. Most of the studies applied a five-point [56,66,79] or seven-point [51,62,65,69,70] Likert scale ranging from "very dissatisfied" to "very satisfied" with a neutral mid-point. Kostianen et al. [47] assessed the occupants' satisfaction with the IAQ on a simplified three-point Likert scale ("very satisfied", "neutral", or "very dissatisfied").

The odour intensity, perceived by the occupants, was measured in eight studies. Four out of eight studies evaluated the odour intensity on a six-point scale ranging from "no odour" to "overpowering/overwhelming/extremely strong odour" [20,49,63,81]. Xie et al. [72] applied a simplified four-point scale ("strong odour", "moderate odour", "weak odour", or "no odour"). Similarly, Alonso et al. [85] applied a four-point scale to assess odour intensity in classrooms. Markowicz and Larsson [52] and Cablé et al. [54] measured the odour intensity on a continuous scale ranging from "no odour" to "extremely annoying odour" and "very comfortable" to "very uncomfortable", respectively. The scale points used by Cablé et al. [54] were not related to odour intensity, but the question posed before the scale relates to the perception of odours, i.e., "Is there any unpleasant smell".

Finally, five studies evaluated occupants' subjective IAQ by measuring their perception of air freshness. Korsavi et al. [67] and Xie et al. [72] let occupants rate their perception of the air freshness on a five-point scale with a neutral mid-point and a four-point Likert scale, respectively. The authors of the first study also asked occupants' air freshness preferences on a binary scale ("fresher" or "as it is") [67]. Jung and Awad [71] assessed occupants' perceived air freshness on a continuous scale ranging from "very stale" to "very fresh". Ma et al. [66] combined an assessment of the air freshness and the level of perceived odours into one four-point Likert scale ("fresh", "stale", "stale with occasional odour", or "stale with odour").

### 3.5. Analysis Methodologies and Modelling Techniques

A majority of the papers used a statistical analysis methodology to investigate relationships between predictor variables (e.g., room temperature, relative humidity, and gender) and the outcome variable (e.g., air quality acceptability, odour intensity, and IAQ satisfaction). The selection of the statistical technique was based on the goal of the analysis, the data type and the data distribution of both the predictor and outcome variables. Three categories of statistical techniques were mainly applied in the reviewed papers, i.e., (1) comparison tests, (2) correlation analysis, and (3) regression analysis. Comparison tests, such as a (paired) *t*-test [62,71], Kruskal–Wallis test [56,65], Mann–Whitney U test [49,50,52,56,61,69,81], and Friedman’s test [64], were applied in studies aiming to determine if there are statistically significant differences in subjective IAQ assessments between different groups (e.g., control and test group, pre- and post-intervention group, ...). A *t*-test is only used when the data have a normal distribution; otherwise, the non-parametric variant (i.e., Mann–Whitney U test) is applied. The *t*-test and Mann–Whitney U test are applicable to determine the difference between two groups. When the number of groups is higher than two, a Kruskal–Wallis test is applied. For studies with a repeated measures design (e.g., studies in which participants’ perceptions are assessed on three or more occasions), a Friedman’s test is used.

Multiple studies performed a correlation analysis to quantify the relationship between an environmental, personal, or contextual variable with subjective IAQ. Various types of correlation coefficients were used, i.e., Pearson correlation coefficient [45,72], Spearman correlation coefficient [67,69,76,83,86], Pearson’s Chi-squared test [59,62,84], and Fisher’s exact test [59,60,84]. One study, by Choe et al. [75], performed a correlation analysis but did not specify the type of correlation coefficient used. The Pearson correlation coefficient is suitable for continuous variables that follow a normal distribution [91]. In case of other data types, such as ordinal variables and non-normally distributed data, a Spearman correlation coefficient is a more appropriate correlation coefficient [91]. Therefore, the Pearson and Spearman correlation coefficients are more suitable when the subjective IAQ is assessed on a continuous or Likert scale, respectively. A Pearson’s Chi-squared test can be applied to determine the relationship between two categorical variables [92]. In the case of a very low sample size, Fisher’s exact test can be performed [60].

Both the comparison and correlation tests give insight into the relationship of one variable (i.e., environmental, personal, or contextual) with subjective IAQ. A regression analysis can account for multiple variables in one regression equation. Therefore, the associations of multiple independent variables on subjective IAQ can be investigated. Multiple types of regression analyses were applied in the reviewed papers, such as linear regression [20,44,57,58,63,66,67,70,77,80,82,85], logistic regression [43,44,46,48,79], ordinal regression [80], and mixed-effects regression models [50,51,54,55,68,78]. A linear regression model is applied when the dependent variable (i.e., subjective IAQ) is a continuous variable. When the dependent variable is a binary (i.e., comprising two categories) or an unordered categorical variable, a logistic regression is more suitable. A logistic regression determines an association between one or more independent variables and the probability of a category of the dependent variable (e.g., satisfied or dissatisfied). In case the dependent variable is an ordered categorical variable (e.g., satisfied or dissatisfied), an ordinal regression should be applied. A linear, logistic, and ordinal regression assume that all datapoints in the investigated dataset are independent. However, this assumption is often violated, e.g., in a longitudinal study when the same occupant is surveyed multiple times, and thus multiple datapoints correspond to the same participant, or when occupants from multiple classrooms or offices are surveyed, and thus occupants and datapoints can be grouped per classroom or office. In cases of dependency in the dataset, a mixed-effects regression



analysis can be performed. A mixed-effects model expands upon a linear, logistic, or ordinal regression model by adding a random effect, which tackles the clustering or hierarchy in the dataset [93]. In other words, the random effect can account for the grouping in the dataset based on the participants or the room. One study applied a hierarchical Bayesian modelling technique to describe occupants' subjective IAQ [47] from a hierarchical dataset. This statistical technique allows prior knowledge to be incorporated into the model [94], making it suitable for analysing complex causal structures.

A regression analysis quantifies the direct associations between an independent and dependent variable (e.g., how variable A affects subjective IAQ). However, there could also be indirect associations that affect a dependent variable (e.g., variable B affects variable A, which affects the dependent variable). To investigate these associations, a latent change model (LCM) or structural equation model (SEM) can be used. Both model types facilitate more complex statistical relationships by including latent variables, which can be inferred from other variables that can be directly observed or measured. An LCM focuses on changes in latent variables over time, while an SEM is not limited to latent variables changing over time. In the current review, LCM [58] and SEM [74] were only applied in one study each. The first quantified the impact of students' change in socio-emotional difficulties and perceived student–teacher relations on their subjective IAQ [58]. The latter applied an SEM to investigate how the concentration of nanoparticles affects occupants' subjective IAQ and how subjective IAQ, in turn, impacts occupants' stress symptoms and risk of perceiving nanoparticles [74]. However, no statistically significant relationship was found between the concentration of nanoparticles and occupants' subjective IAQ.

Only three studies applied a machine learning algorithm that aimed to predict subjective IAQ based on a set of predictors. Gnecco et al. [87] defined Gaussian Naïve Bayes (GNB) models to determine the probabilistic relationships between occupants' subjective IAQ assessment and measured environmental variables. In the studies of Cheung et al. [57] and Kong et al. [73], a black-box modelling approach was employed: the former applied a Gaussian process (GP), while the latter applied a gradient boosting machine (GBM). Such modelling approaches can achieve high prediction accuracies but lack transparency on the relationships between input and output variables. As a result, both studies could not establish an interpretable association between the predictor variables and subjective IAQ. However, the impact of the presence of the predictor variables can be quantified by comparing the prediction accuracies of models with a different number of predictors [57] or by performing a variable importance analysis [73]. The limited possibilities for interpreting the relationships between predictors and outcome variables are one of the drawbacks of black-box modelling approaches such as GP and GBM. However, model interpretability is of less importance when the main goal of the model is to maximise its predictive performance. When the aim of the modelling is to establish relationships, including their magnitude, between variables and subjective IAQ, other approaches such as regression or correlation analysis are more suitable.

### 3.6. Variables Associated with Subjective IAQ

Each of the reviewed studies in Table 4 investigated the associations between one or multiple variables and the subjective assessment of IAQ. Figure 3 provides an overview of the number of studies that investigated associations between a certain variable and subjective IAQ (e.g., air quality acceptability, air freshness, IAQ satisfaction, ...), with the number of studies shown on the y-axis and the different variables on the x-axis. The associations are categorised into the following:

- No association: the relationship between the variable and subjective IAQ was examined but found to be statistically insignificant or negligible in effect size.

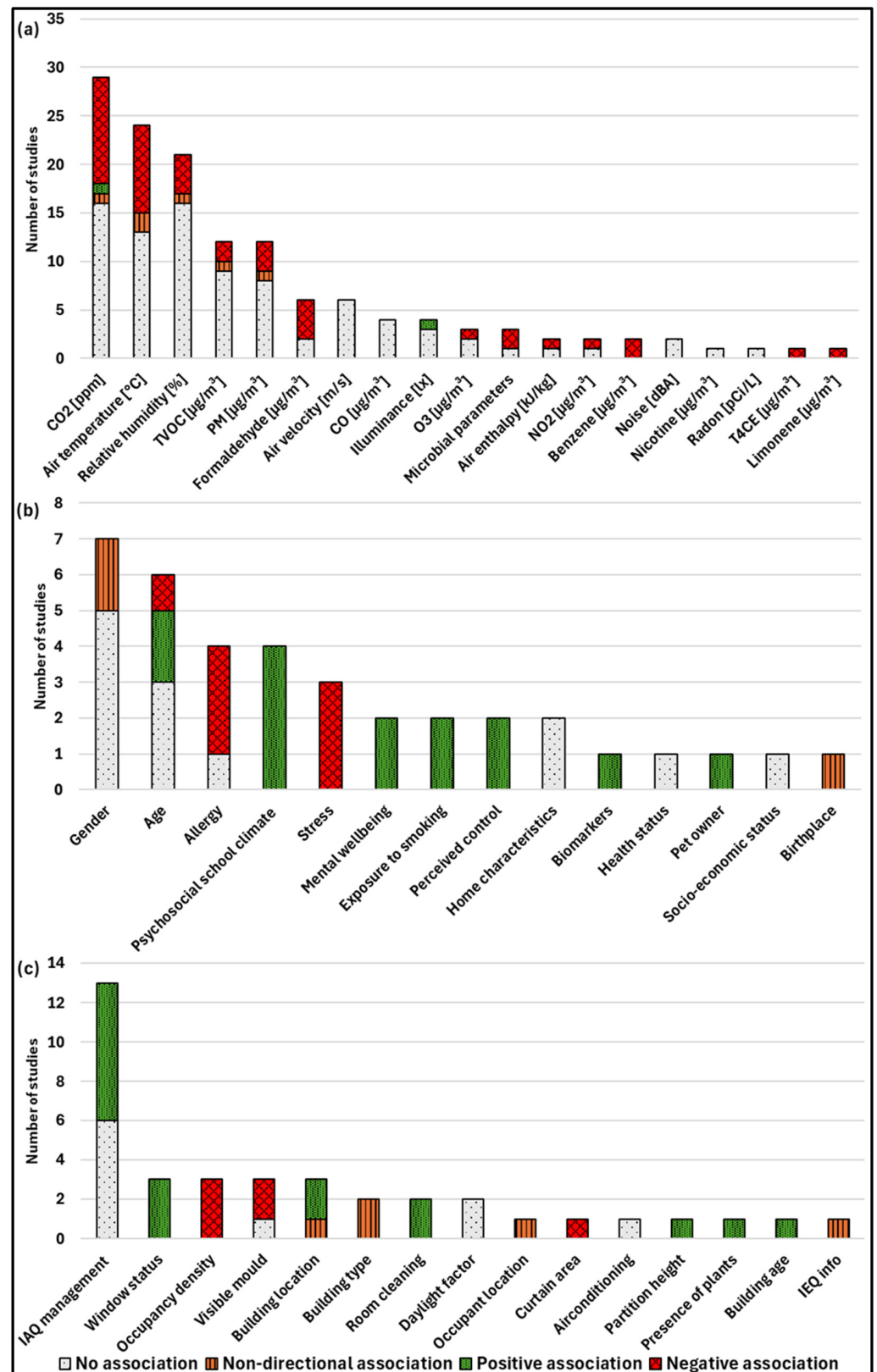


- Non-directional association: significant associations for which no direction (i.e., positive or negative) could be established, typically due to the nature of the variable (e.g., binary or categorical) or the use of black-box models.
- Positive association: an increase in the variable is associated with an improvement in occupants' subjective IAQ, and a decrease corresponds with a decline.
- Negative association: an increase in the variable is associated with a decline in occupants' subjective IAQ, and a decrease corresponds with an improvement.

The results are presented per variable category, i.e., IEQ-related, personal, and contextual variables.

### 3.6.1. Associations with Indoor Environmental Variables

Figure 3 shows that environmental variables were the most frequently investigated in the reviewed studies. CO<sub>2</sub>, indoor temperature, and relative humidity were tested in 29, 24, and 21 studies, respectively, and were, therefore, the most often investigated variables. However, only 24–46% of the studies determined a significant association between the three variables and subjective assessments: CO<sub>2</sub> (13 studies), indoor temperature (11 studies), and relative humidity (5 studies). The majority of the studies indicated a significant negative association between CO<sub>2</sub> and subjective IAQ. However, Kuurola et al. [78] found a positive relationship between CO<sub>2</sub> and subjective IAQ, indicating that an increase in CO<sub>2</sub> concentration would improve occupants' subjective IAQ assessment. The authors [78] stated that the counterintuitive results could be due to the lower ventilation rates in the classroom, which prevented the relative humidity from becoming too low. Sixteen studies found that the association between CO<sub>2</sub> concentration and subjective IAQ was insignificant. Similarly, thirteen studies established an insignificant association between the subjective IAQ and indoor temperature. Of the eleven studies that established a significant association, good agreement was found in the direction of the association between indoor temperature and subjective IAQ. Nearly all of these eleven studies showed a negative association between both, which indicated that an increase in indoor temperature led to a decrease in subjective IAQ assessment. The study by Wu and Wagner [80] established an inverted U relationship between indoor temperature and subjective IAQ, which showed a positive and negative association with indoor temperature for temperatures below and above 20.1 °C, respectively. Similar to indoor temperature, a majority of the papers determined a negative relationship between relative humidity and subjective IAQ, which implied that an increase in relative humidity decreased occupants' IAQ satisfaction in humid climates. The majority of the studies (16 studies) could not determine a significant association between relative humidity and subjective IAQ. Air enthalpy, which combines air temperature and humidity, was investigated in two studies, of which one determined a negative association with subjective IAQ assessments [20]. Six studies analysed the relationship between air velocity and subjective IAQ, but none of the studies determined a significant result between the two.



**Figure 3.** Overview of the number of studies establishing associations of variables with subjective IAQ for (a) IEQ-related variables, (b) personal variables, and (c) contextual variables.

Moreover, various objective IAQ-related variables were studied, such as particulate matter (PM), total volatile organic compounds (TVOCs), formaldehyde, O<sub>3</sub>, CO, benzene, NO<sub>2</sub>, pinene, limonene, tetrachloroethylene (T4CE), nicotine, radon, and mi-

crobial parameters [cells/mg]. Ten of these objective IAQ-related variables were tested by Chatzidiakou et al. [51], i.e., PM, TVOCs, formaldehyde, O<sub>3</sub>, benzene, NO<sub>2</sub>, pinene, limonene, T4CE, and microbial parameters. All the variables showed a significant negative association with subjective IAQ, which indicated that a higher concentration of these pollutants decreased the subjective IAQ assessment of occupants [51]. A negative relationship for PM with subjective IAQ was also found by Park et al. [62] and Smedje et al. [44]. Shim et al. [76] established a negative association between benzene and subjective IAQ. Furthermore, three additional studies [43,76,82] concluded that an increase in formaldehyde concentration decreased the subjective IAQ evaluation. Moreover, Sørensen and Kristensen [83] also established a negative association between TVOC concentrations and subjective IAQ. None of the studies established a significant impact of nicotine, radon, and CO on subjective IAQ.

Four studies investigated the relationship between illuminance and subjective IAQ [43,44,50,79]. One study by Wang et al. [50] established a positive association between a classroom's illumination and pupils' IAQ perception. Finally, the effect of noise was tested in two studies [75,79]. However, no significant relationship between noise and subjective IAQ was determined.

### 3.6.2. Associations with Personal Variables

The impact of personal variables on subjective IAQ assessment has been less investigated in comparison to environmental variables (see Figure 3). The association between gender and subjective IAQ was studied in seven papers and is the most studied personal variable. However, only Kang et al. [56] and Finell et al. [55] found a statistically significant association, indicating that females tended to be more dissatisfied with the IAQ in comparison to males. A majority of the studies did not find a significant association between gender and subjective IAQ. Occupants' age was also tested in six studies, of which three [44,50,51] determined that age had a significant impact on subjective IAQ. The remaining three studies could not establish a significant relationship between age and subjective IAQ. Wang et al. [50] and Chatzidiakou et al. [51] both fitted mixed-effects models including pupils' age. The first established a negative association between age and subjective IAQ, and the latter showed the opposite, that older children were more satisfied with the IAQ than younger children. However, the age range of the tested children was narrow in both studies, i.e., 7 to 13 years old [50] and 9 to 11 years old [51]. Smedje et al. [44] established a positive association between age and subjective IAQ in a sample with a wider age range from 16 to 65 years old. Moreover, Haghighat and Donnini [45] as well as Torriani et al. [77] tested the impact of perceived control on subjective IAQ, and both studies concluded that increasing occupants' perception of control improves their subjective IAQ evaluation. Furthermore, one study by Kang et al. [56] found a significant relationship between occupants' birthplace and their subjective IAQ. However, the study was limited to comparing occupants born in the south of China to occupants born in the north of the country. Another study found that occupants who have pets at their home perceive the IAQ as better compared to occupants who do not own pets [44]. Other characteristics of occupants' homes (e.g., age of the home, number of kids at home, recent repainting at home, . . .) were tested in two studies but were not found to significantly impact people's subjective IAQ assessment [44,50]. Finally, Finell et al. [55] tested the relationship between students' socio-economic status and their subjective IAQ assessment, but no significant association could be determined.

The reviewed studies also investigated the relationship between subjective IAQ and emotional variables, such as mental wellbeing, stress, and psychosocial school climate (i.e., how students feel about their school and their relationship with teachers). Emotional vari-

ables showed a significant relationship in all studies in which these variables were tested. The studies had the same conclusion regarding emotional variables, i.e., (1) occupants' experiencing stress were shown to be more dissatisfied with the IAQ [51,55,65], (2) increasing occupants' mental wellbeing, increased occupants' IAQ satisfaction [47,58], and (3) students having a positive attitude towards their psychosocial school climate (e.g., a good relationship with teachers) were more satisfied with the IAQ in their classroom [44,51,55,58].

Finally, the studies tested health-related variables, such as allergy, biomarkers, exposure to smoke, and health status. Allergy was tested in four studies [44,47,48,50], of which three [44,47,50] found a significant relationship with subjective IAQ, showing that occupants with an allergic background perceived the air as stale. Biomarkers were tested by Bakke et al. [48], who showed that MCA2, a measure for nasal patency, was positively associated with occupants' subjective IAQ evaluation. Exposure to smoke was tested by Smedje et al. [44] and Chatzidiakou et al. [51]. Both studies concluded that people exposed to smoke at home were more likely to report a higher IAQ satisfaction at school. Self-reported health status was tested in one study [54], but this did not lead to a significant relationship with occupants' subjective IAQ.

### 3.6.3. Associations with Contextual Variables

Like personal variables, contextual variables were less often investigated in the reviewed studies in comparison to environmental variables. The IAQ management technique was most often tested in the reviewed studies. IAQ management technique group interventions related to the ventilation system design and operations, such as ventilation rate and air distribution systems. Seven studies identified significant relationships between an IAQ management technique and subjective IAQ. Firstly, Norbäck et al. [49] and Smedje et al. [44] showed that pupils perceive the IAQ in their classroom as better when it was equipped with a displacement ventilation system instead of a mixing ventilation system or an exhaust-only ventilation system. Secondly, Wang et al. [50] and Chatzidiakou et al. [51] found a positive association between pupils' IAQ satisfaction and the air change rate in the classroom. Thirdly, Markowicz and Larsson [52] equipped classrooms with a surface emission trap device (i.e., an absorption cloth) and found that teachers perceived the air as better in those classrooms compared to the classrooms without the device. In addition, Park et al. [62] determined that employees were more satisfied with the IAQ when their office was equipped with a dedicated exhaust in the kitchen and copy area of the office and when the number of people per return air diffuser unit was reduced. Finally, Galičič et al. [84] showed that manually airing classrooms during cleaning resulted in a more favourable subjective IAQ assessment.

Contextual variables, other than IAQ management technique, were only investigated in one to three of the reviewed studies. Except for the operation of the air-conditioning (i.e., on or off) and the daylight factor (i.e., the ratio between indoor and outdoor light level), all remaining contextual variables tested in the reviewed papers were found to have a significant impact on subjective IAQ. Two studies concluded that occupants who were working in a green-labelled office building were more satisfied with the IAQ than occupants working in a non-green-labelled office [53,69]. Furthermore, decreasing the occupancy density in a space did improve the subjective IAQ in three of the reviewed papers [51,63,86]. Multiple building and space-related variables, such as the absence of visible mould [51,68], the presence of operable windows [51,62], a larger distance from traffic [51,84], a smaller curtain area [51], an older building age [44], a higher partition height [62], and the presence of plants [71], were shown to improve occupants' IAQ satisfaction. Moreover, Chatzidiakou et al. [51] found a significant relationship between the timing of cleaning activities in schools and pupils' subjective IAQ assessment. Pupils

were more satisfied with the IAQ when the school was cleaned at the end of the day instead of the start of the day. Kong et al. [73] included the level of information on the IEQ conditions provided to the occupants as a predictor in their machine learning algorithm to predict occupants' subjective IAQ. Finally, the reviewed studies indicated that locating occupants closer to windows and doors [56], and increasing occupants' sense of control over the indoor environment [77] increased their satisfaction with the IAQ.

## 4. Discussion and Outlook

### 4.1. Scope of the Subjective IAQ Analysis

Most of the reviewed studies focused on assessing subjective IAQ and combined this with an often-limited exploration into the main factors influencing the subjective IAQ. The number of variables that were tested, and for which a significant association with subjective IAQ was determined in the reviewed studies, is shown in Section 3.6. In total, 48 different variables were tested in the reviewed studies, of which 38 were identified to have a significant association with subjective IAQ in at least one study. Except for CO<sub>2</sub> (29 studies), indoor temperature (24 studies), humidity (21 studies), PM (12 studies), TVOCs (12 studies), air velocity (6 studies), formaldehyde (6 studies), gender (7 studies), age (6 studies), and IAQ management technique (13 studies), all variables were tested in less than 5 studies, showing that only ten of the 48 variables were frequently investigated in the field-study campaigns. Moreover, the scope of the studies was also limited, with a median of only four different variables that were tested per study. Furthermore, 63% of the studies limited the analysis to only one category of variables, i.e., IEQ-related (19 studies), personal (4 studies), or contextual variables (6 studies). A total of 28% of the papers investigated two categories of variables, of which seven studies include both environmental and contextual factors. The combinations of environmental and personal variables and contextual and personal variables were examined in three studies. Only 4 out of 46 studies tested IEQ-related, personal, and contextual variables together. The limited scope in the number of tested variables makes it clear that current field studies do not aim at defining well-established causal relationships for subjective IAQ, which would aid in a better understanding of occupants' subjective IAQ assessment in real-life buildings. Several studies limited their analysis to quantifying the impact of an intervention (e.g., moving to a green-certified building, changing ventilation systems, etc.) or the change in one environmental variable on the occupants' perception of the IEQ, including occupants' subjective IAQ. Future research should focus on defining general-applicable causal structures for subjective IAQ. Therefore, the combination of multiple IEQ-related, personal, and contextual variables should be studied. Studying the combined effects of multiple variables on subjective IAQ can expose potential interactions between variables, which remain underexplored in the current literature.

### 4.2. Validity of the Statistical Results

Most of the reviewed studies applied techniques and modelling approaches for statistical inference purposes, i.e., determining relationships and causal structures of subjective IAQ. However, two concerns on the validity of these statistical results can be identified, i.e., on the (1) suitability of the analysis technique and (2) sample size. As shown in Section 3.5, a large variation in analysis and modelling techniques was applied in the 46 reviewed studies. Depending on the characteristics of the study and data (e.g., continuous or categorical, repeated-measures design, sample size, ...) and the goal of the analysis (e.g., determining causal structures, pre- and post-tests, quantifying correlation, ...), a suitable statistical technique should be chosen. However, the reviewed studies did not always elaborate on the reasons for choosing a certain analysis technique. As a result, some analysis techniques



seemed unsuitable for the given dataset and analysis goal. Six studies applied a linear regression to determine the relationship between a set of predictors and the subjective IAQ, assessed on a categorical scale [58,63,66–68,70]. However, a logistic or ordinal regression is more suitable in case the dependent variable (i.e., subjective IAQ) is a categorical or ordinal variable. A study by Favero et al. [95] showed that using a linear regression for an ordinal dependent variable causes errors in the statistical relationships. The authors encouraged researchers to apply an ordinal regression model when determining relationships between predictor variables and an ordinal dependent variable [95].

Another issue with the validity of the statistical results is the lack of checking the required sample sizes in the studies. Consequently, it is unclear whether the results obtained have sufficient statistical power. As mentioned in Section 3.3, only 3 [45,69,87] out of 46 studies quantified a required sample size, while the remaining studies only discuss the need for a larger sample size to generalise the obtained results or do not reflect upon the used sample size at all. Moreover, none of the reviewed studies performed a power analysis to quantify the minimum sample size. Nonetheless, open-source software such as G\*Power [96] is available to perform power analyses. The absence of a clear justification of statistical methods used and a quantification of the required sample size via power analysis are common issues in statistical inference studies in the field of IEQ. A literature review on multi-domain comfort studies by Chinazzo et al. [97] showed that 60% of the papers did not justify the choice for a certain statistical method, and only 4% of the studies provided a power analysis to determine or test a required sample size. Moreover, Lan et al. [98] checked 30 studies investigating statistical relationships between IEQ and occupants' productivity, and none of the studies performed a power analysis.

#### 4.3. Diversity of the Studies

As shown in Table 5, a large variation in assessment scales for subjective IAQ was applied in the studies. In total, 46 different scales were identified, which could be grouped into five categories, i.e., (1) IAQ complaint frequency, (2) air quality acceptability, (3) IAQ satisfaction, (4) odour intensity, and (5) air freshness. The differences between the scales could be as minor as differences in labels, such as in [20,81], where both six-point scales were used to assess odour intensity but with slightly different wording. However, often completely different types of scales (e.g., continuous, categorical, or binary scale) were applied across different studies to assess the same variables (e.g., air quality acceptability, air freshness, and odour intensity). Furthermore, a wide range of statistical methods and modelling techniques were used to quantify the associations between predictor variables and subjective IAQ. In total, 20 statistical techniques or modelling algorithms were applied across the reviewed studies, ranging from a simple *t*-test to more complex modelling techniques, such as structural equation modelling and mixed-effects models. This lack of uniformity in evaluation scales and analysis methods made it impossible to compare results of different studies and to perform a meta-analysis of the established associations between predictor variables (i.e., IEQ-related, personal, and contextual factors) and subjective IAQ in the reviewed studies.

#### 4.4. Implications for the Design and Control of HVAC Systems

The practical implications of the defined associations between IEQ-related, personal, or contextual variables and subjective IAQ were often not clarified by the authors of the reviewed studies. However, the established relationships in the reviewed studies could inform improved interior design guidelines and HVAC system design and control strategies. In the case of contextual variables, design guidelines for buildings and HVAC systems could be inferred from the established associations. Examples from this literature



review are limiting certain areas [51], equipping rooms with displacement ventilation instead of mixing ventilation [49], and lowering occupancy densities [51,63] to improve occupants' subjective IAQ assessment. The impact of contextual variables on occupants' subjective IAQ assessment highlights the importance of considering occupant preferences during the building design phase. Further research could investigate an improved method to systematically include subjective IAQ requirements during a building design phase. Moreover, the associations found for IEQ-related and personal variables could benefit HVAC control strategies that account for occupants' subjective IAQ. The determined relationships between IEQ-related variables and subjective IAQ give insights into the environmental parameters of interest for HVAC control systems. This literature review indicated that not only IAQ-related variables, such as CO<sub>2</sub>, TVOC, or PM concentrations, affected occupants' subjective IAQ in field studies, but also thermal-related variables (e.g., indoor temperature, humidity, or air enthalpy). This is in line with various experimental studies showing the effect of thermal-related variables on IAQ perception [99–101]. One study by Yang et al. [20] demonstrated a control strategy that considered the impact of thermal-related variables on IAQ perception. The DCV system of Yang et al. [20] applied a higher CO<sub>2</sub> setpoint during lower indoor temperatures without decreasing the occupants' subjective IAQ evaluation. The determined relationships between personal variables and subjective IAQ, such as age [44,50,51], mental wellbeing [47,58], and stress [51,55,65], indicated the influence of occupants' personal characteristics and background on their perception of the IAQ. This diversity in occupants' subjective IAQ suggests that occupants could perceive the IAQ differently, despite being exposed to the same IEQ and contextual conditions. As a result, the HVAC systems should consider the individual differences in subjective IAQ among occupants in an occupant-centric control (OCC) loop. However, an OCC requires input on preferred IAQ conditions for it to be responsive to occupants' subjective assessment. Currently, an appropriate modelling approach must be defined to translate occupants' feedback on their subjective IAQ to a control input. In contrast to subjective IAQ, various modelling techniques have been defined and tested to integrate occupants' thermal comfort preferences in occupant-centric HVAC control systems, ranging from regression models [102] to machine learning algorithms [103–106]. In the current review, only three studies [57,73,87] aimed at constructing a model that maximises the accuracy of predicting occupants' subjective IAQ. Future research should focus on the use of field measurement data to construct an accurate predictive model for subjective IAQ, which is suitable for integration in an OCC loop.

#### 4.5. Recommendations for Future Research

This section summarises the insights from this review paper and proposes directions for future studies on subjective IAQ assessments in field studies:

- **Focus on personalised modelling approaches for subjective IAQ prediction:** Current literature lacks accurate predictive models for subjective IAQ. Given the significant influence of personal variables showcased in multiple studies, personalised modelling approaches that predict individual IAQ preferences could be promising. Such models may also enhance the performance of occupant-centric HVAC control systems. Therefore, future research should focus on determining suitable modelling approaches which maximise the accuracy of predicting subjective IAQ. In order to investigate personalised modelling approaches for subjective IAQ prediction, more longitudinal data collection campaigns in which occupants' subjective IAQ is assessed multiple times during the measurement campaign are required. The benefit of such a data collection approach is that the dataset consists of multiple subjective IAQ assessments per occupant, making it possible to train a predictive model in learning individuals'

IAQ preferences. Therefore, future studies should focus on longitudinal data collection approaches that lead to datasets with a high number of IAQ assessments per occupant.

- **Investigate the impact of underexplored IAQ-related variables:** This review highlights that the impact of several IAQ pollutants (e.g., benzene, T4CE, and O<sub>3</sub>) on subjective IAQ remains underexplored. Notably, pollutants such as benzene and T4CE consistently showed significant effects in studies where they were examined, unlike conventional pollutants (e.g., CO<sub>2</sub>, TVOCs, and PM), which often exhibited insignificant associations. Future field studies should prioritise these underrepresented pollutants.
- **Ensure statistical validity of research results:** Many of the reviewed papers aimed to establish statistical relationships between IEQ-related, personal, and contextual variables and subjective IAQ assessments. However, a large proportion of these papers failed to clarify the statistical validity of their approach. Future studies should determine minimum sample sizes to ensure adequate statistical power (i.e., 80%) and apply a statistical method that is suitable for the data type that is analysed (e.g., ordinal or continuous variables) and the study design (e.g., accounting for repeated measures). To determine the sample size which ensures sufficient statistical power, researchers could apply open-source software such as G\*Power [96] to perform power analyses.
- **Account for contextual and personal variables:** This review showcased that many personal (e.g., exposure to smoking, exposure to stress, and being a pet owner) and contextual variables (e.g., building location, occupancy density, and presence of plants) have a significant impact on subjective IAQ. These characteristics should be systematically reported, as they may act as confounding variables. Transparent reporting of participant characteristics and building context will help explain variability in subjective IAQ findings across different samples and settings. In addition, the impact of the contextual and personal variables on the predictive performance of models predicting subjective IAQ should be studied, as these could be important predictors.

## 5. Conclusions

This paper presented a systematic literature review on field measurement studies on subjective IAQ evaluations in non-residential buildings. A total of 46 papers were analysed on (1) their context (including geographical location, climate, room type, and ventilation system of the case study), (2) study and survey type, (3) dataset and sample size, (4) subjective IAQ assessment scales, (5) analysis and modelling techniques, and (6) associated variables. This review highlighted current trends and research gaps in the field of subjective IAQ assessment. Based on the identified research gaps, recommendations for future research directions were provided. The main insights of this review are summarised below:

- The vast majority of the reviewed studies (42 out of the 46) were conducted in classrooms or offices. Other non-residential buildings, such as hospitals, sports facilities, and shopping malls, were examined in only one study each and are, therefore, underrepresented. Further research is needed to determine the impact of variables on subjective IAQ assessment in environments other than classrooms and offices.
- A total of 46 different scales to evaluate occupants' perceptions of the IAQ and 20 distinct analysis techniques to determine associations with subjective IAQ were identified from the current review. The lack of a unified assessment and analysis approach for subjective IAQ in field studies made a meta-analysis of the established associations with subjective IAQ unattainable.
- **The validity of the statistical results was often unclear. To guarantee the statistical validity, studies should report a minimum sample size that guarantees sufficient statistical power. Furthermore, studies should elaborate on the suitability of their statistical method for their dataset and analysis goal.**

- The extent of the subjective IAQ analysis in current field studies was, in general, limited. More than half of the reviewed studies investigated the effect of only one to four variables. Moreover, 63% of the studies restricted the analysis to one category of variables (i.e., IEQ-related, personal, or contextual). Future field studies should expand their scope by investigating the combined effect of multiple variables on subjective IAQ in order to define well-established causal structures for subjective IAQ.
- Current field studies focused on a limited exploration of the statistical relationships between a narrow set of IEQ-related, personal, or contextual variables and subjective IAQ. However, only 3 out of 46 studies investigated predictive models for subjective IAQ and quantified the predictive performance of the models. Future research should focus on determining an appropriate and accurate personalised modelling approach for predicting occupants' subjective IAQ assessment in order to include it in an occupant-centric HVAC controller.

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## Abbreviations

The following abbreviations are used in this manuscript:

|        |                                                                            |
|--------|----------------------------------------------------------------------------|
| IEQ    | Indoor environmental quality                                               |
| IAQ    | Indoor air quality                                                         |
| ASHRAE | American Society of Heating, Refrigerating and Air- Conditioning Engineers |
| OCC    | Occupant-centric control                                                   |
| HVAC   | Heating, ventilation and air-conditioning                                  |
| DCV    | Demand-controlled ventilation                                              |
| PAQ    | Perceived air quality                                                      |
| PRISMA | Preferred Reporting Items for Systematic Reviews and Meta-Analyses         |
| RHRN   | Right-here-right-now                                                       |
| NV     | Natural ventilation                                                        |
| MV     | Mechanical ventilation                                                     |
| VAV    | Variable air volume                                                        |
| CAV    | Constant air volume                                                        |
| MVHR   | Mechanical ventilation with heat recovery                                  |
| LogR   | Logistic regression                                                        |
| LinR   | Linear regression                                                          |
| Corr   | Correlation analysis                                                       |
| OrdR   | Ordinal regression                                                         |
| ME     | Mixed-effects                                                              |
| GP     | Gaussian process                                                           |
| LCM    | Latent change model                                                        |
| GBM    | Gradient boosting machine                                                  |

|                 |                                                               |
|-----------------|---------------------------------------------------------------|
| SEM             | Structural equation model                                     |
| GNB             | Gaussian Naïve Bayes probability model                        |
| T               | Indoor temperature [°C]                                       |
| RH              | Relative humidity [%]                                         |
| HCHO            | Formaldehyde [ $\mu\text{g}/\text{m}^3$ ]                     |
| CO <sub>2</sub> | Carbon dioxide [ppm]                                          |
| O <sub>3</sub>  | Ozone [ $\mu\text{g}/\text{m}^3$ ]                            |
| PM              | Particulate matter [ $\mu\text{g}/\text{m}^3$ ]               |
| TVOCs           | Total volatile organic compounds [ $\mu\text{g}/\text{m}^3$ ] |
| T4CE            | Tetrachlorethylene [ $\mu\text{g}/\text{m}^3$ ]               |
| CO              | Carbon monoxide [ $\mu\text{g}/\text{m}^3$ ]                  |
| NO <sub>2</sub> | Nitrogen dioxide [ $\mu\text{g}/\text{m}^3$ ]                 |

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